

Decision-Level Sensor Fusion on the CAN Bus for Autonomous Vehicles: A Standardized Evaluation of Seven Methods

Abstract

Purpose – To evaluate decision-level sensor fusion on the CAN bus via a standardized workflow, identifying models balancing accuracy, responsiveness, safety, and robustness.

Design/methodology/approach – Seven methods (Rule-Based, Bayesian, Dempster-Shafer, XGBoost, SmallMLP, Temporal Sequence, Hybrid) are evaluated deterministically across 15 scenarios using synchronized LiDAR, camera, and ultrasonic inputs. Performance is assessed under baseline, moderate, and severe sensing-stress (noise, packet loss, jitter). Results are aggregated across five randomised seeds with bootstrap confidence intervals. Metrics include F1-score, mean time-to-detect (MTTD), Full_Brake false-negative rate (FNR), and near-miss rate (NMR).

Findings – SmallMLP achieves the highest baseline F1-score (0.7824), whilst XGBoost exhibits the strongest robustness under degraded sensing. Rule-Based and Dempster-Shafer remain closest to the Expected operating point, indicating the strongest benchmark alignment, whereas the Hybrid model provides a balanced safety-oriented compromise via deterministic overrides.

Originality – A CAN-synchronized sensor fusion benchmarking framework enforces identical sensing, actuation, and logging. Ensuring strict control-plane parity enables a fair comparison of heterogeneous decision-level fusion architectures.

Practical implications – Guides systems engineers in selecting auditable decision modules for safety-critical systems based on operational constraints, safety requirements, and robustness objectives.

Research limitations/implications – Evaluation is simulation-based and restricted to LiDAR, camera, and ultrasonic modalities. Real-world validation remains future work.

Keywords CAN bus; autonomous vehicles; decision-level fusion; sensor fusion benchmarking; safety-critical systems; XGBoost; Dempster-Shafer theory; mean time-to-detect; false-negative rate.

Paper type Research paper.

1. Introduction

Autonomous vehicle (AV) decision-making relies heavily on robust sensor fusion, yet evaluating heterogeneous decision logics, ranging from deterministic rules to neural networks, remains a significant challenge. Prior work often optimizes isolated stacks, such as perception or intrusion detection, or mixes sensors without a plug-compatible decision interface. This leaves message-level reliability under-explored and head-to-head tests rare. While conflict-aware fusion with mitigation has begun to address Controller Area Network (CAN) security (1), a standardized benchmark comparing rule-based, probabilistic, evidential, learning-based, temporal, and hybrid decision modules under identical CAN timing is still missing. Recent studies on decision-level fusion, multimodal perception, and late-fusion driver-state estimation each contribute useful advances in their own settings, including real-time LiDAR-camera decision fusion for vehicle detection, 4D radar-camera benchmarks for all-weather perception, and multi-sensor late fusion for cognitive-state monitoring (2–4). Together, these studies highlight the diversity of modern fusion approaches and motivate a reproducible CAN-synchronized benchmark for comparing decision modules under identical timing and actuation conditions.

To address this gap, this study enforces deterministic scheduling, identical actuator mapping, and reproducible logging to benchmark heterogeneous decision logics head-to-head. Motivated by time-triggered, endpoint-based interfaces that enable predictable delays and strict synchronization on CAN-like networks (5), we provide control-plane parity. Seven decision modules, Rule-Based (RB), Bayesian (BF), Dempster-Shafer (DS), XGBoost (XGB), SmallMLP (SMLP), Temporal Sequence (TS), and Hybrid (HB), consume synchronized LiDAR, dual front cameras, and rear ultrasonic inputs to emit one of four CAN-encoded actions. The shared CAN-synchronized pipeline enables direct comparison of F1, Full_Brake false-negative rate (FNR), and mean time-to-detect (MTTD) across 15 scripted scenarios. This

design isolates decision logic from timing variability, enabling fair comparison under identical inputs and evaluation conditions.

2. Related Works

Endpoint-based, time-triggered CAN interfaces enable deterministic latency and synchronization, supporting controlled comparisons of decision-fusion methods under fixed update rates (5). Recent work on runtime monitoring and vehicular health diagnostics further emphasizes reproducible evaluation, heterogeneous data integration, and transparent AI operation in autonomous systems (6,7). High-dimensional late-fusion approaches target scene understanding, whereas this study focuses on decision-level fusion under controlled timing and actuation conditions (2–4).

In security-sensitive CAN environments, multi-detector aggregation utilizes feature ranking, autoencoders, and LSTM-based estimation to manage conflict. This illustrates how evidential and decision-level fusion enhance resilience against adversarial or anomalous traffic (1). Recent ITS work also combines Dempster-Shafer fusion with adversarial deep learning to improve resilience against attacks, reinforcing evidential reasoning as a relevant baseline for security-sensitive transportation systems (8). Explainability remains central in safety-critical systems, motivating transparent rule-based reasoning, fuzzy decision support, and evidential fusion methods capable of handling uncertainty and conflicting sensor observations (9–12).

Learning-based fusion and hybrid designs extend this line of work by improving predictive capability while preserving compactness. Multimodal RGB and event fusion with attention improves lateral control prediction (13), while lean, CAN-integrated AV prototypes demonstrate cost-effective stacks with limited sensor suites (14). Recent studies also demonstrate that lightweight tree ensembles and multilayer perceptrons perform well on structured, low-dimensional signals, motivating the evaluation of XGBoost and compact MLP-style models for decision fusion (15–17). Recent studies further demonstrate the benefits of multimodal sensing, robust perception, and explainable federated learning for improving reliability, uncertainty tolerance, and interpretability in autonomous systems (18,19).

A recent Internet of Vehicles (IoV) study adds an important adjacent direction: federated learning and blockchain can support privacy-preserving distributed vehicular intelligence, secure model updates, and efficient traffic rerouting in edge-cloud environments (20). However, that line of work addresses distributed traffic intelligence rather than a plug-compatible, decision-level benchmark under identical CAN timing. Despite these advances, the literature lacks a standardized low-dimensional comparison of RB, BF, DS, XGB, SMLP, TS, and HB under identical actuator mapping, seed control, stress conditions, and safety-focused metrics.

3. Method

This paper presents a CAN-bus framework for evaluating seven decision-level fusion methods under identical timing, actuation, and logging.

3.1 Simulation Framework and Pipeline

The simulation pipeline (Fig. 1) evaluates all fusion algorithms through initialization, simulation, and post-processing.

3.2 System Initialization

Before each run, the framework initializes the decision module, virtual CAN devices, evaluation metrics, and monitoring dashboard.

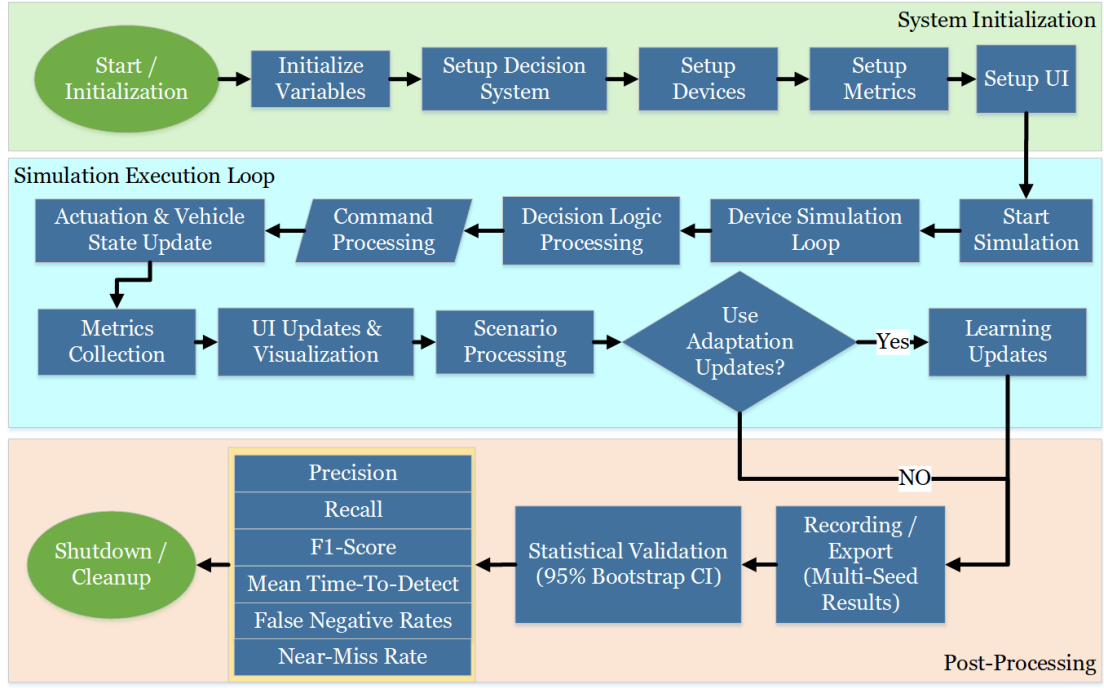


Figure 1. Simulation workflow for CAN bus decision-level fusion pipeline

Default parameters include $speed_0 = 60.0$ km/h (initial vehicle speed), $rpm_0 = 4000$ (engine rotation), $f^{Lidar} = 4$ Hz, and $f^{ATmega} = 20$ Hz (update frequencies). The decision core is expressed as

$$DecisionCore = A(\theta) \quad (1)$$

where denotes the selected inference method and the model parameters or weights. The virtual CAN network is defined by

$$devices = \{s_i = (CAN_ID_i, f_i, state_i)\}_{i=1}^N \quad (2)$$

where each sensor node represents a LiDAR, camera, or ultrasonic device with an identifier, update rate, and current state.

$$l = \sqrt{\frac{6}{n_{in} + n_{out}}} \quad (3)$$

where and are the numbers of input and output neurons, respectively; non-neural models are loaded from stored files.

3.3 Core Fusion Logic: Implementation of Evaluated Methods

3.3.1 Rule-Based Fusion (RB)

The rule-based fusion model deterministically maps sensor states to discrete actions (Eq. 4).

$$D = \{NO_Action, Slowdown, Partial_Brake, Full_Brake\} \quad (4)$$

At each time step, sensor inputs consist of the LiDAR label $L \in D$, camera occupancy flags F^L and $F^R \in \{Free, Not\ Free\}$, and the rear ultrasonic distance $B > 0$ cm, which is active only for $B < 400$ cm (parking range). Decisions follow a fixed-priority sequence in which the

LiDAR output governs the primary action, camera observations refine the decision, and ultrasonic sensing provides the final adjustment.

$$D_1 = \begin{cases} Full_Brake, & L = Full_Brake, \\ Partial_Brake, & L = Partial_Brake, \\ Slowdown, & L = Slowdown, \\ NO_Action, & otherwise, \end{cases} \quad (5)$$

camera cues refine the outcome as

$$D_2 = \begin{cases} Partial_Brake, & \mathbf{1}_{(F_L = Not_Free) \vee (F_R = Not_Free)} = 1 \text{ and } D_1 = NO_Action, \\ D_1, & otherwise, \end{cases} \quad (6)$$

and ultrasonic sensing adjusts it via

$$D_3 = \begin{cases} Full_Brake, & B < 50 \text{ and } D_2 = NO_Action, \\ Slowdown, & 50 \leq B \leq 400 \text{ and } D_2 = NO_Action, \\ D_2, & otherwise. \end{cases} \quad (7)$$

The final output is determined by LiDAR priority, followed by cameras and ultrasonic sensing, and each action is mapped to a command byte $D_{final} = D_3$

$$c(D_{final}) \in \{0x00, 0x01, 0x02, 0x03\} \quad (8)$$

Boolean actuator flags are derived directly from the final decision, ensuring deterministic behaviour.

3.3.2 Bayesian Fusion (BF)

The Bayesian fusion model defines probabilistic reasoning over the discrete action set in (4). Each action begins with a prior probability. At each decision step, the combined likelihood from all sensors is computed as

$$L(d) = \prod_{s \in S} \lambda_s(d) \quad (10)$$

where $\lambda_s(d)$ denotes the likelihood weight contributed by sensor s , with LiDAR dominating (e.g., $\lambda_{\text{lidar}}(Full_Brake) = 0.95$) and ultrasonic input active only for short distances. The unnormalized posterior is

$$\tilde{P}(d) = P(d) \max[L(d), 10^{-10}] \quad (11)$$

and its normalized form becomes

$$P(d | E) = \frac{\tilde{P}(d)}{\sum_{d' \in \mathcal{D}} \tilde{P}(d')} \quad (12)$$

where E represents the current sensor evidence. The final decision is the maximum a posteriori estimate

$$\hat{d} = \arg \max_{d \in \mathcal{D}} P(d | E) \quad (13)$$

A hysteresis rule suppresses rapid oscillation before the final action is encoded as a control byte.

3.3.3 Dempster-Shafer Fusion (DS)

The Dempster-Shafer fusion model performs evidential reasoning over the frame of discernment

$$\Theta = \{ \text{NO_Action}, \text{Slowdown}, \text{Partial_Brake}, \text{Full_Brake} \} \quad (14)$$

where each sensor s defines a basic probability assignment $m_s: 2^\Theta \rightarrow [0,1]$. Sensor reliability is adjusted by Shafer's discounting

$$m'_s(A) = \alpha_s m_s(A) \quad (15)$$

$$m'_s(\Theta) = \alpha_s m_s(\Theta) + (1 - \alpha_s) \quad (16)$$

where $\alpha_s \in [0,1]$ denotes the sensor confidence factor. For two sensors, the conjunctive combination computes the unnormalized mass

$$\tilde{m}(A) = \sum_{B \cap C = A} m_1(B) m_2(C) \quad (17)$$

and the conflict coefficient

$$K = \sum_{B \cap C = \emptyset} m_1(B) m_2(C) \quad (18)$$

where $K \in [0,1]$ measures disagreement. Normalized mass is

$$m(A) = \frac{\tilde{m}(A)}{1-K}, A \neq \emptyset \quad (19)$$

where $K < 1$. When $K = 1$, complete conflict occurs and a predefined safety fallback is activated. The fused evidence m_{final} is converted to probability via the pignistic transform

$$\text{BetP}(\theta) = \sum_{A \ni \theta} \frac{m_{\text{final}}(A)}{|A|}, \theta \in \Theta \quad (20)$$

where $\text{BetP}(\theta)$ denotes belief for hypothesis θ . The final action is

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \text{BetP}(\theta) \quad (21)$$

representing the most probable decision under uncertainty, with safety overrides applied when needed (e.g., LiDAR "Full Brake").

3.3.4 XGBoost Fusion (XGB)

The XGBoost fusion model encodes each temporal sensor snapshot into a feature vector that combines binary states and normalised continuous values:

$$x_t = [\tilde{a}, \mathbf{1}[L_{\text{free}}], \tilde{B}, \tilde{v}, \tilde{e}, \mathbf{1}[S_{\text{flag}}]]^T \quad (22)$$

where $\tilde{a}$ represents the normalised acceleration, L_{free} denotes the binary lane-free state (utilising the indicator function $\mathbf{1}[\cdot]$ to map to $\{0,1\}$), $\tilde{B}$ is the normalised rear ultrasonic distance, $\tilde{v}$ is the vehicle speed, $\tilde{e}$ is the engine rate, and S_{flag} represents the critical safety boundary flag. The model is trained as a multiclass probabilistic classifier using gradient-boosted decision trees.

$$F_c(x_t) = \sum_{k=1}^K f_k(x_t) \quad (23)$$

These raw scores are subsequently converted into normalised class probabilities using the softmax function:

$$P(y = c|x_t) = \frac{\exp(F_c(x_t))}{\sum_{j=1}^C \exp(F_j(x_t))} \quad (24)$$

where $P(y = c|x_t)$ denotes the predicted probability of executing action class c , and C represents the total number of discrete driving action classes. The optimal baseline decision $\hat{y}_{base}$ is obtained by selecting the class with the maximum posterior probability:

$$\hat{y}_{base} = \arg \max_{c \in C} P(y = c | x_t) \quad (25)$$

This baseline decision is then refined by a temporal smoothing filter and a deterministic safety boundary function, $\text{SafetyGuard}(\cdot)$, to produce the final decision $\hat{y}_{final}$.

3.3.5 SmallMLP Fusion (SMLP)

The SmallMLP maps a temporal sensor window into four driving actions. Each input sample is structured as a feature vector concatenating normalised CAN-bus readings into a single sequence. The architecture consists of a standard two-layer feedforward neural network comprising h hidden units equipped with ReLU activation functions, followed by a final dense layer utilising a softmax activation to project the logits into class probabilities.

The network is trained with minibatch gradient descent and L2 regularization, and its output is mapped to the actuator commands in Equation (8).

3.3.6 Temporal Sequence Fusion (TS)

The Temporal Sequence (TS) fusion model captures short-term dependencies within a rolling window of sensor readings. The use of 1-D convolutions rather than recurrent neural network (RNN) or Long Short-Term Memory (LSTM) architectures is deliberate, as CAN-bus events are locally correlated in time and short kernels (typically 3 to 5 frames with stride 1) can efficiently extract these patterns without recurrent overhead. The resulting convolution outputs are aggregated into a temporal representation and projected through a linear classifier to generate softmax action probabilities.

3.3.7 Hybrid Fusion (HB)

The Hybrid Fusion model combines deep neural inference with deterministic, rule-based safety gating.

$$x_t = [\tilde{a}, \mathbf{1}[L_{free}], \tilde{B}, \tilde{v}, \tilde{e}, \mathbf{1}[S_{flag}]]^T \quad (26)$$

where $\tilde{a}$ is acceleration, L_{free} is the lane-free state, $\tilde{B}$ is rear distance, $\tilde{v}$ is vehicle speed, $\tilde{e}$ is engine rate, and S_{flag} is a critical safety flag. The inference stage uses a lightweight two-layer multilayer perceptron (MLP) to estimate the probability distribution over the four driving actions, after which a deterministic safety arbitration layer applies explicit safety constraints before issuing the final decision.

3.4 Fusion Methods Comparison

Table I summarizes the feature implementation methods of each model.

Table I. Feature Implementation Methods by Model Type
[Insert Table I Here]

3.5 Post-Processing

3.5.1 Parameter Metrics

After each simulation run, the framework evaluates each fusion model using accuracy, reliability, and responsiveness metrics. For each class $c \in D = \{\text{NO_Action}, \text{Slowdown}, \text{Partial_Brake}, \text{Full_Brake}\}$, the confusion matrix elements are defined as

$$TP_c = |\{t: d_t = c \wedge g_t = c\}| \quad (27)$$

$$FP_c = |\{t: d_t = c \wedge g_t \neq c\}| \quad (28)$$

$$FN_c = |\{t: d_t \neq c \wedge g_t = c\}| \quad (29)$$

$$TN_c = |\{t: d_t \neq c \wedge g_t \neq c\}| \quad (30)$$

where TP, FP, FN, and TN are the standard confusion-matrix terms.

$$Precision_c = \frac{TP_c}{TP_c + FP_c} \quad (31)$$

$$Recall_c = \frac{TP_c}{TP_c + FN_c} \quad (32)$$

$$F1_c = \frac{2 Precision_c Recall_c}{Precision_c + Recall_c} \quad (33)$$

The Full_Brake class is treated as safety-critical and is evaluated using its false-negative rate (FNR), defined through the event indicators $\mathbf{1}_{\text{pred}=F}(t)$ and $\mathbf{1}_{\text{gt}=F}(t)$:

$$TP_F = \sum_t \mathbf{1}_{\text{pred}=F}(t) \mathbf{1}_{\text{gt}=F}(t) \quad (34)$$

$$FN_F = \sum_t (1 - \mathbf{1}_{\text{pred}=F}(t)) \mathbf{1}_{\text{gt}=F}(t) \quad (35)$$

$$FNR_F = \frac{FN_F}{P_f} \quad (36)$$

$$P_f = TP_F + FN_F \quad (37)$$

where TP_F and FN_F denote true positives and false negatives, respectively, for the Full_Brake class, FNR_F is the Full_Brake false-negative rate, and P_f is the total number of ground-truth Full_Brake events. Temporal responsiveness is measured by the Mean Time-To-Detect (MTTD),

$$\mu_{\text{MTTD}} = \frac{1}{n} \sum_{i=1}^n t_{\text{detect},i} \quad (38)$$

$$\sigma_{\text{MTTD}} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (t_{\text{detect},i} - \mu_{\text{MTTD}})^2} \quad (39)$$

where μ_{MTTD} is the mean detection time and σ_{MTTD} is its variation; median, quartiles, and extrema capture response variability.

3.5.2 Data Recording and Export

The simulator logs synchronized sensor/state/decision streams and exports raw and aggregated CSVs with metadata for reproducible comparison.

3.5.3 Expected Operating Point and Baseline Equilibrium

Because all algorithmic evaluations share identical scenario distributions and ground-truth action frequencies across five randomised seeds, the system establishes an Expected operating point, for example, an Expected F1-score of 0.7333.

The Expected operating point represents the deterministic reference behaviour of the benchmark under nominal sensing and traffic conditions. Rather than serving as a maximum-performance target, it defines the baseline equilibrium against which different fusion methods are compared. In the subsequent performance analysis, this reference serves as the primary comparative anchor for assessing behavioural consistency across models. Algorithmic robustness is therefore evaluated through both conventional performance metrics and the

degree of deviation from the Expected operating point under degraded sensing conditions. Consequently, higher metric values do not necessarily indicate better benchmark alignment if they reflect a larger departure from the Expected reference.

4. Results and Discussion

The full set of tabular results is available as supplementary data on Zenodo (DOI: 10.5281/zenodo.20717304).

4.1 Experimental Setup

Seven decision modules (Rule-Based: RB, Bayesian: BF, Dempster-Shafer: DS, XGBoost: XGB, SmallMLP: SMLP, Temporal Sequence: TS, and Hybrid: HB) were evaluated in a deterministic CAN-bus simulation across 15 scenarios. All models were executed on the exact same set of five randomised seeds, and results are reported as means with 95% confidence intervals and standard deviations. Performance and robustness are evaluated relative to the Expected operating point. The summary tables are generated from per-seed CSV outputs.

4.2 Functional Performance Analysis

4.2.1 Classification Performance

Table II summarizes the baseline functional performance of all evaluated models relative to the Expected operating point. Under baseline conditions, RB and DS most closely approximate the Expected operating point ($F1 = 0.7333$), both achieving $F1 = 0.7324$. XGB (0.7892) and SMLP (0.7824) achieve the highest raw classification performance but deviate further from the reference behaviour. HB provides a compromise between behavioural consistency and classification capacity, whereas BF and TS show weaker alignment with the Expected operating point.

Table II. Baseline Functional Performance Relative to the Expected Operating Point
[Insert Table II Here]

4.2.2 Decision-Specific Performance

Table III summarizes the F1-score performance of each model across four scenario families. Scenario-level performance is evaluated relative to the Expected operating point rather than raw F1-score alone. Parking (S1) strongly penalizes Bayesian, while RB and DS remain closely aligned with the Expected operating point. Gradual motion (S2 to S4) compresses performance differences, whereas sudden obstacles (S5) and complex trajectories (S6) amplify model divergence. RB and DS exhibit highly similar scenario-family F1 patterns, despite differences in latency, robustness, and safety metrics. While XGB, SMLP, and HB often achieve higher raw F1-scores, RB and DS remain most consistent with the Expected operating point.

Table III. Scenario-Family F1-Score Relative to the Expected Operating Point
[Insert Table III Here]

4.3 Model-Specific Analysis

Table IV. Safety Reliability and Latency Profiles Relative to the Expected Operating Point
[Insert Table IV Here]

4.3.1 Timing and Latency Conformity

As shown in Table IV, MTTD reveals a distinct speed hierarchy relative to the 0.000 s reference point. XGB operates as the fastest learned model (0.0210 s), while RB, DS, and BF cluster around a short, stable latency near 0.05 s. HB is moderately fast at 0.2225 s, SMLP is slower at 0.7153 s, and TS is operationally the least responsive at 1.1694 s. Learned models provide strong classification performance, but responsiveness varies substantially across architectures.

4.3.2 Safety Balance Preservation

XGB, SMLP, HB, DS, and RB preserve the expected zero Full_Brake FNR under baseline conditions. BF and TS deviate from this target, suggesting they would likely require secondary safety gating prior to deployment.

4.3.3 Hazard Sensitivity

Near-Miss Rate (NMR) stays tightly grouped for the viable models, with XGB, SMLP, RB, DS, and HB forming a balanced cluster, while BF appears lower primarily due to under-sensitivity rather than superior safety. This supports utilizing FNR and MTTD alongside NMR, rather than evaluating NMR in isolation.

4.4 Cross-Scenario Performance Analysis

Models were evaluated under moderate and severe degradation consisting of sensor noise, latency jitter, and CAN-bus packet loss, as summarized in Table V. Robustness is evaluated through the magnitude of performance drift relative to the Expected operating point. Under severe degradation, XGB exhibits the smallest F1 decline (-0.0798) and minimal latency change (+0.0039 s). SMLP retains strong classification performance but incurs a substantial latency increase (+1.6772 s). BF, DS, and RB remain relatively stable but experience larger performance degradation, while TS deteriorates across both accuracy and responsiveness. The degradation benchmark uses the fixed stress settings baseline (noise_level = 0.00, drop_prob = 0.00, jitter_ms = 0), moderate (0.10, 0.05, 25), and severe (0.20, 0.10, 40), with noise, message drops, and jitter injected consistently across models.

Table V. Robustness Degradation under Simulated Sensor Noise

[Insert Table V Here]

In summary, the robustness benchmark demonstrates that different fusion strategies exhibit distinct operational trade-offs under degraded sensing conditions. Using the Expected operating point as a reference baseline provides a more nuanced assessment of deployment viability than the isolated maximization of any single metric.

5. Conclusions

This study presented a standardized benchmark for decision-level sensor fusion on the CAN bus, utilizing a deterministic workflow to compare seven heterogeneous decision modules under identical inputs, timing, actuation, and logging. By evaluating 15 scripted scenarios across five randomised seeds with bootstrap confidence intervals, the framework enabled a reproducible comparison of rule-based, probabilistic, evidential, learning-based, temporal, and hybrid decision logics against the Expected operating point.

Comparative analysis reveals that no single model dominates across all criteria. RB and DS most closely preserve the expected operating balance under clean conditions; XGB exhibits the most robust performance under degraded sensing conditions; SMLP achieves high baseline classification performance but incurs a larger latency cost; and HB offers a balanced compromise between classification capability and safety preservation. Conversely, BF and TS proved less suitable for deployment within this specific benchmark, exhibiting weaker classification alignment and diminished safety or responsiveness under stress. These findings confirm that decision-level fusion architectures must be evaluated jointly through comprehensive metrics, such as F1-score, Full_Brake FNR, MTTD, and robustness drift, rather than relying on a single performance indicator. Scenario-family analysis further showed that gradual hazards, sudden obstacles, and complex motion favour different fusion strategies, reinforcing the need for multi-metric evaluation across diverse operational conditions.

These results demonstrate that compact, auditable decision modules can be evaluated fairly within a shared CAN-synchronized pipeline, supporting fusion-method selection based on operational priorities. The benchmark also validates deterministic reference behaviour as a comparison anchor while recognizing that some learned models may outperform that reference on conventional metrics.

Despite these contributions, the study has limitations. The evaluation was entirely simulation-based, the sensing configuration was restricted to LiDAR, dual front cameras, and rear ultrasonic inputs, and intermediate action classification remains a source of error. Future work should extend this validation to physical on-road scenarios, integrate additional sensing modalities such as radar or infrared sensors, and investigate data balancing strategies to improve the detection of rare safety-critical events. In addition, further study is needed to determine how temporal context can improve hazard anticipation without introducing the latency and instability observed in short-window temporal fusion models.

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